Section B   Que. 41 to 55

Ans. 41 → Goal of Computer Networks & Applications
Goal - Resource sharing, communication, reliablity and cost reduction.
Application - Email, file sharing, video conferencing --- etc.

Ans - 42 - OSI Reference Model. The OSI Model has 7 layer: Physical, Data link, Networking, Transport, session, Presentation. aplication. It standrdizes Communication between system.

Ans - 43. Transmission modes - ~~Half~~ duplex - Two way, simultaneous Comm
Simplex: one-way Communication.
Full Duplex - Two way, simultaneous Communication.

Ans - 44. ISDN (Integrated Services Digital Networks)
ISDN is a digital communication Network that carries voice, video, and data over telephone.

Ans - 45. Channel Allocation methods.
Static allocation: fixed bandwidth allocation.
Dynamic allocation. Bandwidth allocated as needed, improving efficiency.

Ans - 46 - LAN Protocols: Common LAN Protocols.
Ethernet (IEEE 802.3): uses CSMA/CD
Token Ring (IEEE 802.5): uses token passing

Ans - 47 - Slotted ALOHA
Slotted ALOHA divides time into slots, reducing collisions.
Maximum throughput is 36.8% better than pure ALOHA (18.4%).

Ans 48 - CSMA/CD - Senses the channel before transmitting, detects collision, stops transmission, and retranmits after a random delay.

Ans 49 - function of Data Link layer - Framing
Error detection.
Flow control.
MAC addressing.

Ans 50 - Sliding window protocol.
Allows multiple frames to be sent before acknowledgment improving transmission efficiency and flow control.

Ans 51 - Services of Network layer.
Provides logical addressing (IP), routing, packet forwarding, and Congestion Control.

Ans - 52 - Distance vector Routing Algorithm Routers exchange routing tables with neighbors and select the shortest ~~pasing~~ path using Bellman-Ford algorithm.

Ans - 53 - IP Addressing. - IP Addressing uniquely Identifies devices on a network. It consists of Network ID and host ID.

Ans - 54 - Subnetting and Advantages. Subnetting divides a network into smaller subnetworks. Advantages: Better IP utilization, improved security, reduced congestion.

Ans - 55 - Difference Between Routing and forwarding.
Routing. Determines the best Path for packets.
forwarding. Sends packets to the next hop using the routing table.

Section C Que- 56 to 58

Ans 56 – Design a Network for a university with three ~~Capu~~ Campuses.

Suitable Network Architecture - Hybrid client - Server Architec - ure.

Technologies & Devises Required -

* LAN in each Campus using Ethernet switches.
* WAN Connecting the three campuses using leased Line/MPLS /VPN.
* Centralized Database server at the main campus.
* Routers to connect LANS the WAN
* Firewalls for network security
* Wireless Access Points (WiFi) for mobile devices.
* Manged switches for resource sharing.
* Fiber-optic backbone for high-speed Communication

Requirment Fulfilled:

1 * Resource sharing: - Shared printers, files, and servers over LAN

2 * Centralized Database Access: Database servers ~~over LAN~~ accessible from all Campuses through WAN.

3 * Secure Communication - VPN, Firewall, and encryption (SSL/IPSec).

4 * Internet Connectivity. - Router with ISP Connection Shared across the Network.

5 * Easy Expansion - Modular design with VLANs and Scalable switches/routers.

Ans - 57

To build an efficient and reliable office network, the company should use the following IEEE Standards.

1. Wired LAN inside the office.

* IEEE Standard: IEEE 802.3 (Ethernet)
* Description – IEEE 802.3 defines Ethernet technology used for wired local Area Networks (LANs). it provides high-speed, reliable, and low latency communication between computers, printers, servers and switches using copper cables (Cat 5e / Cat 6 / Cat 6A)
* Advantages :-
  - High data transfer speed (100 Mbps, 1 Gbps, 10 Gbps or more)
  - Reliable communication with low packet loss.
  - Easy installation and maintanace.
  - Cost-effective for office Networking.

2. Wireless Network for employees-

* IEE Standard. (IEE 802.11 ax) (wifi 6)
* Description. (It support High-speed Comunication with improved.)
* Advantages
  - High speed wireless access
  - Supports multiple and efficiency.
  - Secure Communication using WPA3 encryption.

* Fiber-based High speed backbone.

* Summary Table

Requirement	IEEE Standard	Technology.
wired LAN	IEEE 802.3	Ethernet.
Wireless LAN	IEEE 802.11 ax	WiFi 6
Fiber Backbone	IEEE 802.3z / IEEE 802.3ae	Gigabit / 10 Gigabit Ethernet over Fiber.

Conclusion: Using IEEE 802.3 for wired networking. IEEE 802.11 ax for wireless access, and IEEE 802.3z / 802.3ae for the fiber backbone ensures a fast secure, reliable, and scalable office Network.

Section C - Que. 58

Ans- 58 - A company receives the network 192.168.10.0/24. it needs 4 equal subnets. ~~Find~~.

* New subnet Mask → 255.255.255.192 (/26)
* Network addresses → (192.168.10.0/26) (192.168.10.64/26) - - - etc.
* Host range of each subnet.
  * subnet 1 - 192.168.10.1 - 192.168.10.62
  * Subnet 2 - 192.168.10.65 - 192.168.10.126
  * subnet 3 - 192.168.10.129 - 192.168.10.190
  * subnet 4 - 192.168.10.193 - 192.168.10.254

Step 1 New subnet mask -

* Given network 192.168.10.0/24
* Required subnets = 4
* Number of bits to borrow = 2 (because $2^2 = 4$)

New prefix = /26

Subnet mask = 255.255.255.192

Step-2 - Number of Hosts -

Remaining host bits = 6

usable hosts = $2^6 - 2 = 62$

Each subnet supports 62 usable hosts.

Step-3 - Network Addresses and Host Ranges.

Subnet	Network Address	Host Range	Broadcast
Subnet-1	192.168.10.0/26	192.168.10.1 - 192.168.10.62	192.168.10.63
Subnet-2	192.168.10.64/26	192.168.10.65 - 192.168.10.126	192.168.10.127
Subnet-3	192.168.10.128/26	192.168.10.129 - 192.168.10.190	192.168.10.191
Subnet-4	192.168.10.192/126	192.168.10.193 - 192.168.10.254	192.168.10.255

This creates 4 equal subnets, each with 62 Usable host addresses.